

Notes: Dugast & Foucault (JFE, 2018)

Data Abundance and Asset Price Informativeness

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1 Big picture

- **Question.** Does a decline in the cost of accessing data necessarily make asset prices more informative?
- **Main answer.** Not necessarily. More abundant cheap data can increase short-run price informativeness but reduce long-run price informativeness if it crowds out demand for more precise processed information.
- **Core friction.** Information processing takes time. Low-precision raw signals can be obtained quickly; high-precision processed signals arrive later because filtering noise out of raw data is costly and slow.
- **Main model object.** Information sellers produce two signals about the payoff of a risky asset: a raw signal available at date 1 and a processed signal available at date 2.
- **Raw signal.** The raw signal reveals the payoff with probability θ and is pure noise with probability $1 - \theta$.
- **Processed signal.** The processed signal reveals both the raw signal and whether it is informative or noise. It is more precise but arrives later.
- **Information market.** Raw and processed signals have fixed production costs C_r and C_p but can be replicated at zero marginal cost. Competitive information sellers choose fees, and speculators decide which signal to buy.
- **Trading market.** Speculators trade on their signals against liquidity traders and a competitive risk-neutral market maker, as in Kyle-style trading.
- **Key mechanism.** A fall in C_r attracts more traders to the raw signal. This can make date-1 prices reflect the raw signal before processed information arrives.
- **Good case.** If the raw signal is often noise, more raw trading can create mispricing that processed-signal traders later correct. Demand for processed information can rise.
- **Bad case.** If the raw signal is sufficiently reliable, more raw trading reveals information early and reduces the rents from processed information. Demand for processed information can fall or disappear.
- **Main implication.** Big data and FinTech can speed up price discovery in the short run while weakening incentives for deeper fundamental research in the long run.
- **Empirical predictions.** The model predicts changes in analyst coverage, price informativeness horizons, trade covariance across periods, momentum/contrarian patterns, and the relation between raw-data reliability and trading behavior.

2 Model primitives and measurement

2.1 Timing

The model has four dates $t \in \{0, 1, 2, 3\}$.

- At $t = 0$, markets for information open. Sellers decide whether to produce and sell raw and processed signals.
- At $t = 1$, raw-signal traders observe the raw signal s and trade.
- At $t = 2$, processed-signal traders observe (s, u) , where u indicates whether the raw signal is informative, and they trade.
- At $t = 3$, the asset payoff $V \in \{0, 1\}$ is realized.

Interpretation: The model explicitly builds in the fact that quick information is less filtered, while precise information takes time.

2.2 Asset payoff and raw signal

The risky asset payoff is

$$V \in \{0, 1\}, \quad \Pr(V = 1) = \Pr(V = 0) = \frac{1}{2}.$$

The raw signal is

$$\tilde{s} = \tilde{u} \times \tilde{V} + (1 - \tilde{u}) \times \tilde{\varepsilon}, \tag{1}$$

where $\tilde{u} = 1$ with probability θ and $\tilde{\varepsilon}$ is an independent noise draw equal to zero or one with equal probability.

- If $u = 1$, the raw signal is correct: $s = V$.
- If $u = 0$, the raw signal is pure noise and independent of V .
- The parameter θ measures the reliability of the raw signal.

Interpretation: The raw signal can be very useful or completely misleading. The model's central comparative statics depend on how reliable this cheap raw signal is.

2.3 Processed signal

The processed signal is the pair (s, u) .

- It confirms whether the raw signal is informative.
- It arrives at date 2, one period after the raw signal.
- It represents activities such as deeper analysis, noise filtering, natural language processing with validation, or fundamental research.

Interpretation: The processed signal is not simply a more expensive version of the raw signal. It is a delayed signal that tells traders whether the raw signal should be trusted.

2.4 Information production and fees

There are fixed costs:

$$C_r = \text{cost of producing the raw signal}, \quad C_p = \text{cost of producing the processed signal}.$$

Once produced, a signal can be replicated at zero marginal cost. Information markets are competitive, so fees satisfy zero-profit conditions for sellers and zero expected net trading profits for signal buyers.

Interpretation: This Veldkamp-style feature matters because information is nonrival. A cost decline can affect both the amount of information produced and the number of speculators using it.

2.5 Speculator demand for signals

Let

$$\alpha_1 = \text{mass of speculators who buy the raw signal}, \quad \alpha_2 = \text{mass of speculators who buy the processed signal}.$$

The equilibrium demands are α_1^e and α_2^e .

Interpretation: The endogenous variables of interest are not just prices. They include whether the market for processed information survives when raw data become cheap.

2.6 Order flow and prices

At dates 1 and 2, speculators trade one share conditional on their signal. Liquidity traders submit exogenous orders. The market maker observes aggregate order flow f_t and sets price

$$p_t = E[V \mid f_t].$$

Interpretation: Prices become informative when aggregate order flow allows the market maker to infer what informed traders know.

3 Models and theoretical specifications

3.1 Date-1 equilibrium

At date 1, the market maker observes order flow f_1 and sets $p_1 = E[V \mid f_1]$.

- If enough speculators trade on the raw signal, order flow can fully reveal s .
- If too few speculators trade on the raw signal, liquidity trading masks their information.
- The date-1 equilibrium price is therefore either fully revealing or uninformative depending on the mass of raw-signal traders.

Interpretation: Cheap raw information increases α_1 , which can make the first-period price reflect the raw signal before deeper processing arrives.

3.2 Date-2 equilibrium

At date 2, processed-signal traders observe (s, u) and the date-1 price. Their trading depends on whether the raw signal was genuine or noise.

- If $u = 1$, processed traders know the raw signal is correct.
- If $u = 0$, processed traders know that any price movement due to the raw signal was noise-driven.
- They can therefore either reinforce date-1 price discovery or trade against date-1 mispricing.

Interpretation: The processed signal is valuable either because it confirms correct information or because it identifies noisy information that should be reversed.

3.3 Expected profit from processed information

The gross expected profit from the processed signal can be written as

$$\bar{\pi}_2(\alpha_1, \alpha_2) = \alpha_1 \pi_2^c(\alpha_2) + (1 - \alpha_1) \pi_2^{nc}(\alpha_2), \quad (2)$$

where π_2^c is the profit conditional on a date-1 price change and π_2^{nc} is the profit conditional on no date-1 price change.

Interpretation: More raw traders increase the chance that the date-1 price moves. This can either help or hurt processed traders depending on whether the raw signal is noise or information.

3.4 Effect of raw-signal demand on processed-signal value

The derivative of processed-signal value with respect to raw-signal demand can be positive or negative. The paper identifies a threshold $\hat{\alpha}_2(\theta)$ such that increasing α_1 raises the value of processed information when α_2 is below the threshold and reduces it when α_2 is above the threshold.

Interpretation: This is the nonstandard part of the model. More traders using raw signals need not crowd out deeper research if raw signals are sufficiently unreliable; it can create profitable correction opportunities.

3.5 Equilibrium information markets

The equilibrium depends on production costs C_r and C_p .

- If C_r is high, the raw signal is not produced or has low demand.
- If C_r falls, raw-signal demand rises.
- If C_p is high or raw information becomes too informative too early, the processed signal may stop being produced.

Interpretation: The model is about endogenous information production. Price informativeness depends on which information markets survive.

3.6 Price informativeness

The paper measures price informativeness at date t by

$$\xi_t(C_r, C_p) = \frac{1}{4} - E[(V - p_t^*)^2]. \quad (3)$$

- The maximum value is $1/4$, when price fully reveals V .
- The minimum value is zero, when price is uninformative.
- ξ_1 is short-run informativeness and ξ_2 is long-run informativeness.

Interpretation: A fall in raw-data cost can increase ξ_1 while reducing ξ_2 if it crowds out processed information.

3.7 Main comparative static

A reduction in C_r :

1. weakly increases short-run price informativeness;
2. can reduce long-run price informativeness;
3. can eliminate production of the processed signal when the raw signal is sufficiently reliable and cheap.

Interpretation: Big data can make markets faster but not necessarily deeper.

4 Identification logic and empirical predictions

This is a theoretical paper, so the relevant “identification” is the set of empirical patterns that would support or reject the mechanism.

4.1 Prediction 1: analyst coverage and fundamental research

When the cost of raw data falls and the raw signal is sufficiently reliable, demand for processed signals can decline.

Interpretation: Empirically, this predicts that easier access to machine-readable raw information can reduce analyst coverage or demand for costly fundamental research for some stocks.

4.2 Prediction 2: horizon-specific price informativeness

Cheap raw data should improve short-horizon price informativeness. But long-horizon informativeness can fall when processed information is crowded out.

Interpretation: Empirical work should distinguish short-run and long-run informativeness. A single informativeness measure may miss the key tradeoff.

4.3 Prediction 3: reversal after raw-information shocks

If raw signals are often noise, processed traders later correct first-period price movements. This generates reversals and negative covariance between first-period trades and second-period informed trades.

Interpretation: News analytics or social media signals with low reliability should be associated with later corrections.

4.4 Prediction 4: momentum when raw information is reliable

If raw signals are reliable, processed traders confirm them. The second-period return can be positively related to the first-period return.

Interpretation: The same decline in raw-data cost can generate different return patterns depending on raw-signal reliability.

4.5 Prediction 5: trade covariance across periods

The model predicts how the covariance between speculators’ trades at dates 1 and 2 varies with θ and C_r .

Interpretation: This provides a way to test the model using signed order flow and identifiable information events.

5 Tables and figures

The paper is theoretical and figure-intensive. The figures below are directly cropped from the original paper and grouped to save space.

5.1 Figure 1: Timing

Read:

- Raw information is purchased at date 0 and observed at date 1.
- Processed information is also purchased at date 0 but arrives at date 2.
- Speculators trade when their information arrives, and the asset payoff is realized at date 3.

Economic message: The timing is the paper's core modeling innovation: low-precision information is fast, high-precision information is slow.

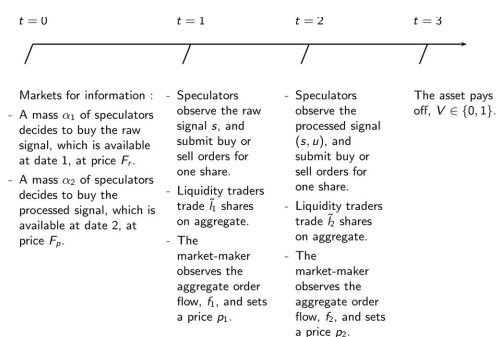


Fig. 1. Timing.

with past returns (see their Proposition 3) while
our model.

equal to the asset payoff while with probability (1 - α)
is just noise.⁸

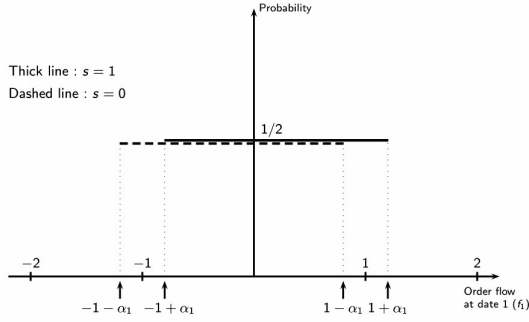
The processed signal is obtained after filtering
noise from the raw data available just before date

5.2 Figure 2 + Figure 3: Equilibrium prices at dates 1 and 2

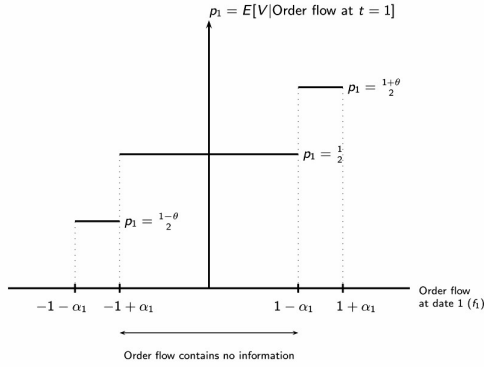
Read:

- Figure 2 shows how date-1 order flow can or cannot reveal the raw signal.
- When raw-signal demand is large enough, order flow moves price to reflect s .
- Figure 3 shows date-2 price paths after processed information arrives.
- If the processed signal confirms the raw signal, price movements can continue in the same direction.
- If the processed signal invalidates the raw signal, date-2 trading corrects the date-1 movement.

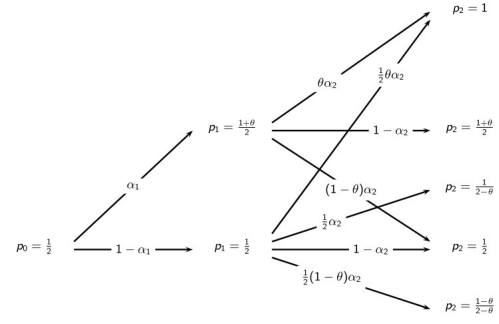
Economic message: The same initial price movement can later become momentum or reversal depending on whether raw information was genuine.



(B) Equilibrium price at date $t = 1$ (p_1)



(a) Fig. 2: date-1 order flow and price.



(B) Equilibrium distribution of the order flow (f_2) at date 2 , when $p_1 = 1/2$, and $\alpha_2 < 1$

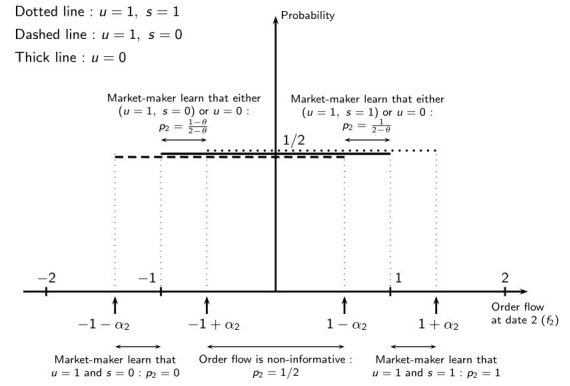


Fig. 3. Equilibrium at date 2. Panel A shows the equilibrium paths for the asset price when $s = 1$ and their likelihood of occurrence. Panel B shows

(b) Fig. 3: date-2 price paths.

5.3 Figure 4 + Figure 5: Value of processed information

Read:

- Figure 4 shows parameter regions in which more raw-signal demand raises or lowers the value of processed information.
- When raw reliability is low and processed-signal demand is not too high, more raw trading can create correction opportunities.
- Figure 5 plots the aggregate gross expected profit from processed information as a function of processed-signal demand.
- The profit curve determines whether the processed-signal market is active.

Economic message: More cheap raw data can either complement or substitute for processed information. This is the key ambiguity in the model.

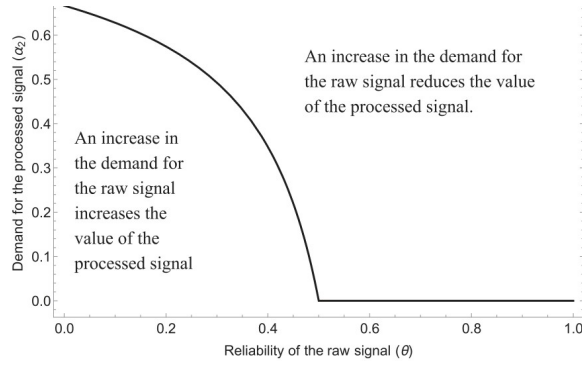
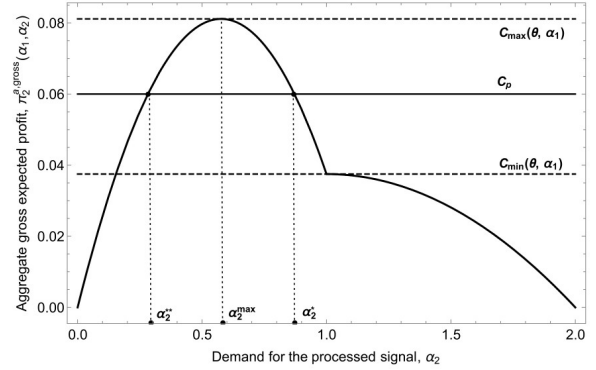


Fig. 4. This figure shows the sets of values of θ and α_2 for which a marginal increase in the demand for the raw signal (α_1) increases or decreases the value of the processed signal for speculators. The curve separating the two regions is the threshold $\hat{\alpha}_2(\theta)$ defined in [Corollary 2](#). All variables are defined in [Appendix A](#).

(a) Fig. 4: regions where raw demand affects processed value.



(b) Fig. 5: profit from processed signal.

5.4 Figure 6 + Figure 7: Equilibrium signal demands and cost regions

Read:

- Figure 6 shows raw-signal demand and processed-signal demand as the raw-signal cost changes.
- As C_r falls, raw-signal demand increases.
- Processed-signal demand first may rise but can then fall sharply or disappear.
- Figure 7 maps the cost regions where the processed signal is produced and where lowering raw-signal cost crowds it out.

Economic message: Cheap raw information can crowd out deeper information production. The effect depends on both θ and the relative costs of the two signals.

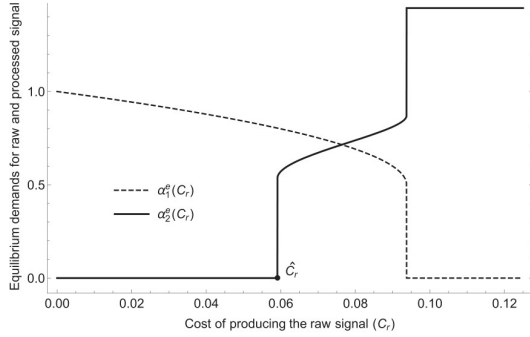
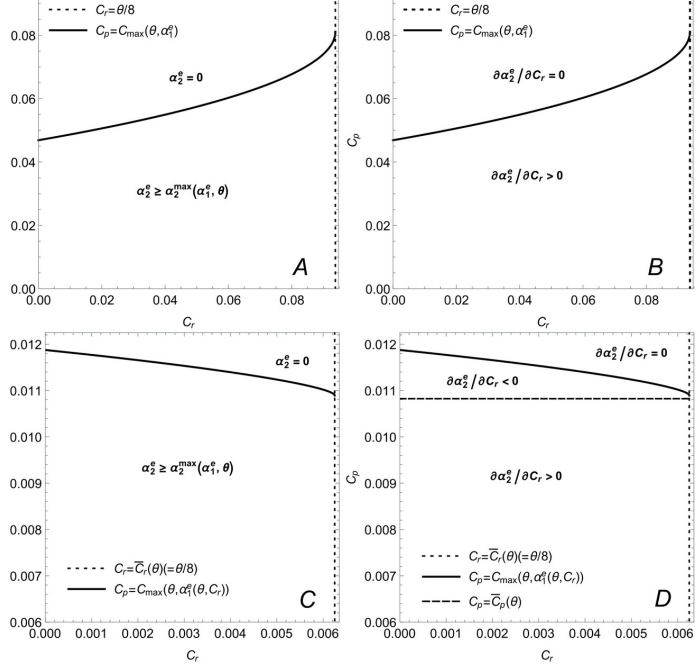


fig. 6. Equilibrium demands for the raw signal (dashed line) and the processed signal (thick line) as a function of the cost of producing the raw signal, C_r (X-axis). Other parameters are $\theta = 0.75$ and $\bar{C}_p = 0.06$. All variables are defined in [Appendix A](#).

(a) Fig. 6: equilibrium demands.



(b) Fig. 7: cost regions and crowding-out zones.

5.5 Figure 8 + Figure 9: Price informativeness

Read:

- Figure 8 shows short-run informativeness ξ_1 and long-run informativeness ξ_2 as functions of raw-signal cost.
- Short-run informativeness rises as raw information becomes cheaper.
- Long-run informativeness can decline when the processed-signal market shrinks or shuts down.
- Figure 9 maps where changes in raw-signal cost raise or lower long-run informativeness.

Economic message: The model overturns the usual intuition that lower information costs always improve informativeness. Faster information can reduce deeper information.

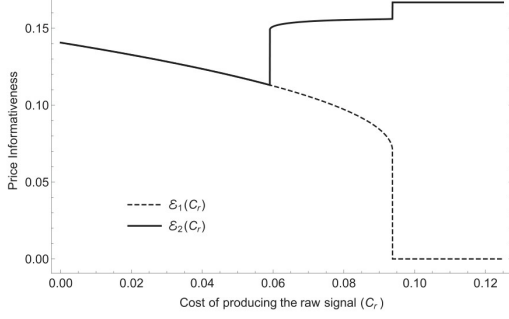
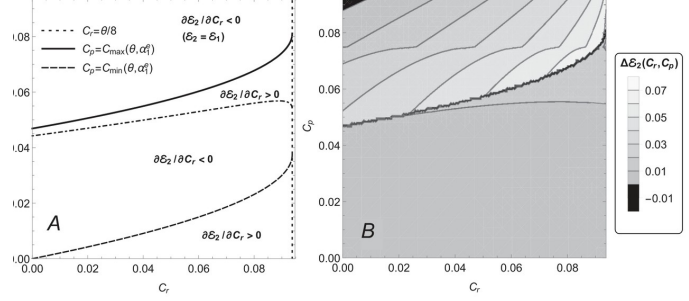


Fig. 8. Price informativeness in the short run (dashed line) and the long run (thick line) as a function of the cost of producing the raw signal C_r (X-axis). Other parameters are $\theta = 0.75$ and $C_p = 0.06$. All variables are defined in [Appendix A](#).

from trading on the processed signal is just equal to the
(a) Fig. 8: short-run versus long-run informativeness.



Panel A indicates for each pair (C_r, C_p) the sign of the derivative of long run price informativeness with respect to the cost of the raw signal, i.e., $\partial \bar{E}_2(C_r, C_p) / \partial C_r$. The dashed line is $C_{\max}(C_p, \alpha_1^*(C_r))$ while the dashed-dotted line is the set of values for C_r and C_p such that $Y(C_r, C_p, \theta) = 0$ (where Y is defined in the online appendix (see Section 5)). Panel B indicates for each pair (C_r, C_p) , the range of values for $\Delta \bar{E}_2(C_r, C_p) = \bar{E}_2(\infty, C_p) - \bar{E}_2(C_r, C_p)$, i.e., the difference between long run price informativeness when producing the raw signal is so costly that nobody acquires it (i.e., $C_r > \frac{1}{\theta}$) and the price informativeness in equilibrium when the raw signal is produced. The raw data corresponds to positive values of $\Delta \bar{E}_2(C_r, C_p)$ and the data correction to negative values.

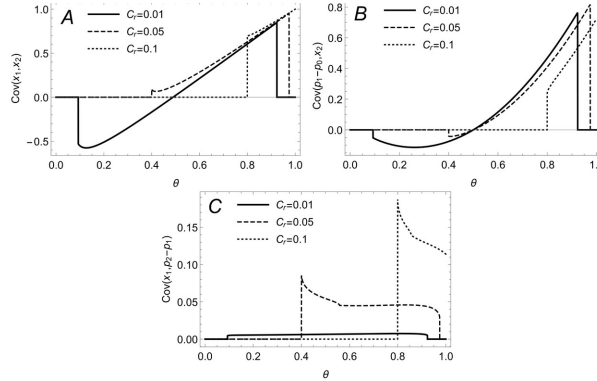
(b) Fig. 9: local effects of raw-data cost.

5.6 Figure 10: Trading covariances

Read:

- Panel A shows covariance between speculators' date-1 and date-2 orders.
- Panel B shows covariance between the first-period return and the date-2 informed order.
- Panel C shows covariance between the first-period speculator trade and the second-period return.
- The signs vary with raw-signal reliability θ and raw-signal cost C_r .

Economic message: The model produces testable predictions for order-flow dynamics: raw-data reliability determines whether later informed trading reinforces or corrects earlier price movements.



Panel A shows the covariance between speculators' orders at dates 1 and 2 ($\text{Cov}(x_1, x_2)$) as a function of θ . Panel B shows the covariance between π in period 1 ($\pi_1 = p_1^* - p_0$) and speculators' orders in period 2 as a function of θ . Panel C shows the covariance between speculators' order and the return in period 2 ($p_2^* - p_1^*$) as a function of θ . In each case, various values of C_r are considered: $C_r = 0.1$ (dotted lines), $C_r = 0.05$ (dashed lines), and $C_r = 0.01$ (solid lines).

5.7 Table 1: Main variables in Appendix A

Read:

- The table summarizes core notation: payoff V , raw signal s , reliability θ , type indicator u , prices, demands, order flow, expected profits, costs, and price informativeness.
- It is useful because the paper is notation-heavy and uses multiple equilibrium objects.

Economic message: Keeping track of the notation clarifies the structure: costs determine signal demands; signal demands determine order flow; order flow determines prices; prices determine informativeness.

r_t	price of the raw signal
α_1	Demand for the raw signal
α_2	Demand for the processed signal
l_t	Demand from liquidity traders for the asset at date t
x_t	Demand of the asset by a speculator at date t
f_t	Order flow at date t
p_t	Price of the asset at date t
Ω_t	Dealers' information set at date t
$\bar{\pi}_1(\alpha_1)$	Expected trading profit from trading on the raw signal
$\bar{\pi}_2(\alpha_1, \alpha_2)$	Expected trading profit from trading on the processed signal
$\pi_2^{nc}(\alpha_2)$	Expected trading profit on the processed signal conditional on no price change at date 1
$\pi_2^c(\alpha_2)$	Expected trading profit on the processed signal conditional on a price change at date 1
C_p	Fixed cost of producing the processed signal
C_r	Fixed cost of producing the raw signal
$\pi_2^{gross,a}(\alpha_1, \alpha_2)$	Aggregate gross expected trading profit from trading on the processed signal
$\bar{\pi}_2^{net}(\alpha_1, \alpha_2)$	Net expected trading profit from trading on the processed signal
$\mathcal{E}_t(C_r, C_p)$	Price informativeness at date $t \in \{1, 2\}$
r_t	Return from date $t - 1$ to date t

¹⁸ When some investors buy the raw and the processed signals, the total cost of information production is $C_r + C_p$ in our model. As C_r declines, this cost goes down. Moreover, if the decline in C_r is sufficiently large, the processed signal stops being produced (see [Proposition 3](#)) so that the total cost of information production is only C_r .

6 Short summary

- The paper studies how cheaper access to raw data affects asset price informativeness.
- Raw signals are fast but noisy; processed signals are slow but precise.
- The raw signal is correct with probability θ and pure noise otherwise.
- A fall in the raw-signal cost attracts more traders to raw information.
- This makes prices more likely to reflect raw information at date 1.
- If raw information is noisy, processed traders can profit by correcting mistakes, so processed-information demand can increase.
- If raw information is sufficiently reliable, early price discovery reduces the value of processed research, so processed-information demand can fall.
- Short-run price informativeness weakly rises when raw-data costs fall.
- Long-run price informativeness can fall if cheap raw data crowds out processed information.
- The paper predicts changes in analyst coverage, trade covariance, momentum, reversal, and horizon-specific informativeness.
- The broad lesson is that data abundance can make prices faster but not necessarily more informative about fundamentals in the long run.

7 Conclusion

7.1 Core conclusion

Dugast and Foucault show that lower raw-data costs do not automatically improve market informativeness. When information processing is time-consuming, cheap raw data can change the incentives to produce deeper signals. The market can become faster while becoming less deeply informed.

7.2 Economic intuition

Cheap raw data draws traders into early, low-precision signals. This can help if those signals are often noise, because processed traders later correct the resulting mispricing. But if raw data are sufficiently reliable, early prices already reveal much of the signal. Processed information then becomes less profitable, and fewer traders buy it. The result is a possible decline in long-run informativeness.

7.3 Takeaway

The paper provides a theory of a modern information market in which big data and FinTech are not unambiguously welfare-improving for price discovery. More data can speed up price changes, but the most informative research may be crowded out if markets rely too heavily on cheap fast signals.